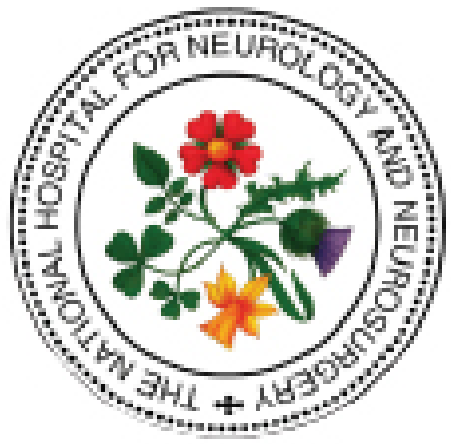




The National Hospital for Neurology and Neurosurgery

Neuroscience Critical Care Unit (NCCU)

CENTRAL NERVOUS SYSTEM (CNS) INFECTION

CNS infections, meningitis, encephalitis, brain abscess, are rare but time-sensitive emergencies – certain features are common to all but speed at which they develop, and intensity varies depending on causative organism – mortality and morbidity are significant.

MENINGITIS Is a regional infection affecting the meninges, usually the arachnoid and pia mater, and the intervening space. Signs and symptoms include fever, headache, neck stiffness, non-blanching haemorrhagic rash in meningococcal meningitis.

Bacterial meningitis occurs when organisms gain access to subarachnoid space either through bacteraemia (from upper airway source), contiguous spread from dental or sinus infections, traumatic or congenital communications with exterior, or a neurosurgical procedure. The severe inflammation associated with bacterial meningitis results in oedema of brain and meninges, and eventually increased ICP once compensatory mechanisms for CSF displacement overwhelmed. Bacterial meningitis is associated with significant morbidity and mortality. Acute bacterial meningitis is a neurological emergency and clinical outcome dependent on immediate antibiotic therapy

Viral Meningitis has no specific anti-viral therapy for most causes and treatment is largely supportive with spontaneous recovery anticipated in most cases.

Fungal Meningitis is almost always a disease of the immunocompromised.

ENCEPHALITIS is a diffuse infection affecting brain tissue +/- inflammation of the meninges. Signs and symptoms include altered mental state (confusion, drowsiness), seizures, changes in personality and behaviour. It is a life-threatening condition with high mortality (as high as 70% if left untreated) and morbidity rates. Early identification and treatment is essential.

BRAIN ABSCESS is a localised area of necrosis with a surrounding membrane within the brain parenchyma, usually from an infectious or traumatic process. The most frequent microbial pathogens isolated are Staphylococcus / Streptococcus.

Signs and symptoms include focal neurology, seizures headache, nausea, vomiting.

TESTS and INVESTIGATIONS – BUT DO NOT DELAY STARTING TREATMENT

All meningitis patients should be isolated for 24 hours unless meningococcal disease excluded

Use Epic '**ADULT MENINGITIS PANEL**' to ensure all recommended investigations sent (type 'Meningitis' into Orders)

- Blood cultures before antibiotics prior to broad spectrum antibiotic / antiviral therapy wherever possible and **WITHIN 1 HOUR OF PRESENTATION**
- Venous blood gas (or consider arterial if hypoxic)
- FBC, U&E, LFTs, CRP, Clotting, Glucose
- Blood for pneumococcal and meningococcal PCR (purple top EDTA)
- HIV test (purple top EDTA)
- Serum save (gold top)

CT scan (If CT shows brain abscess with mass effect surgery essential)

Lumbar puncture safe if:

LP in PATIENTS on ANTICOAGULANTS

- ✓ not in coma
- ✓ no focal neurological signs
- ✓ no papilloedema

CSF analysis – for subsequent definitive treatment

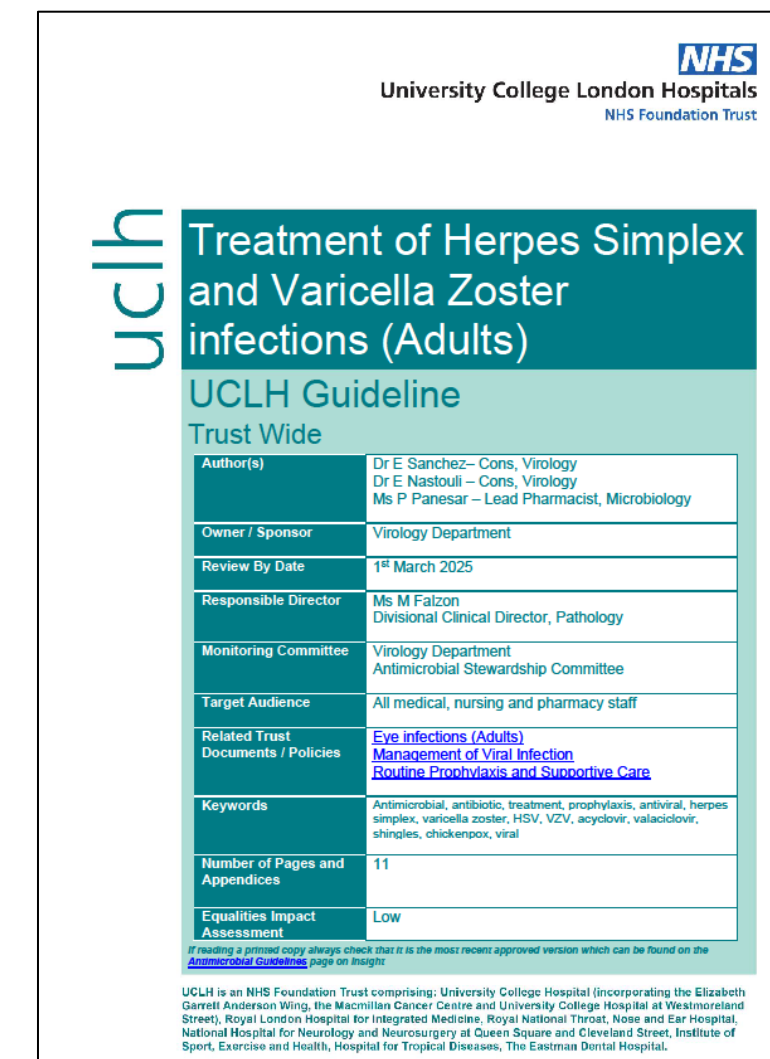
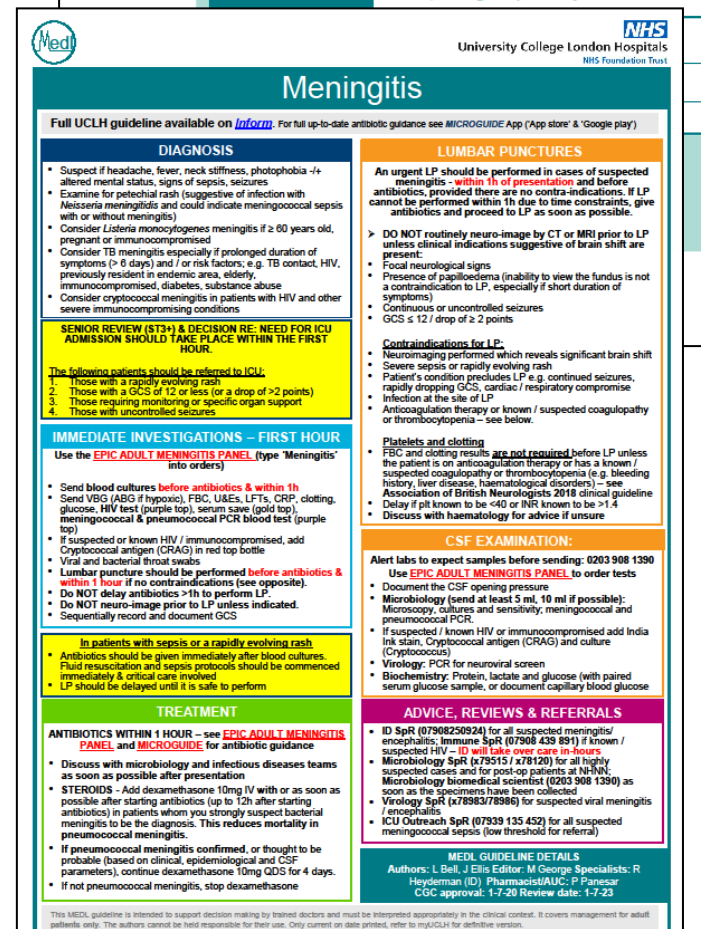
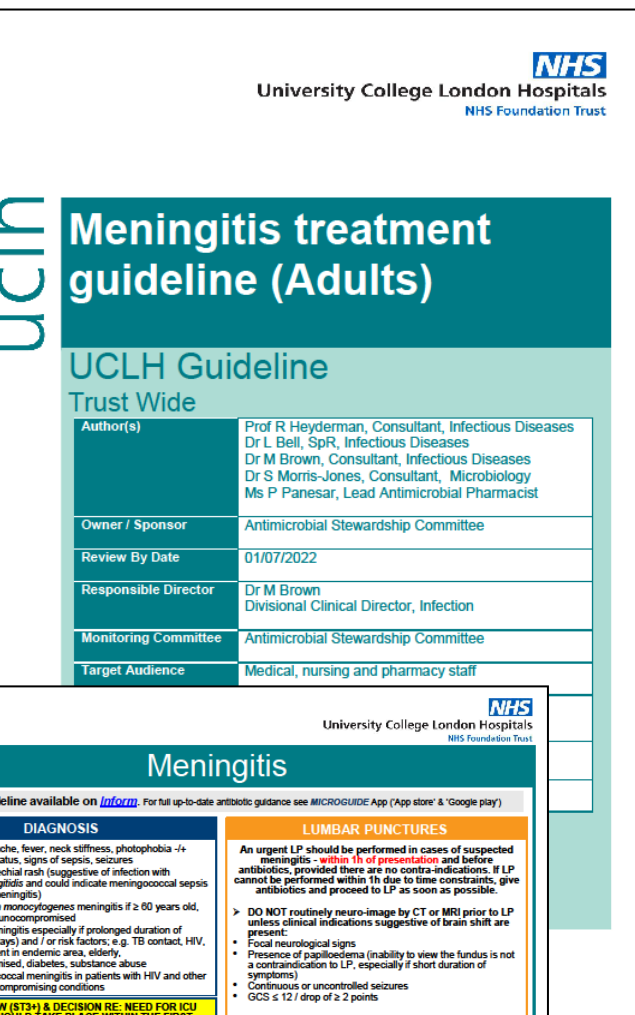
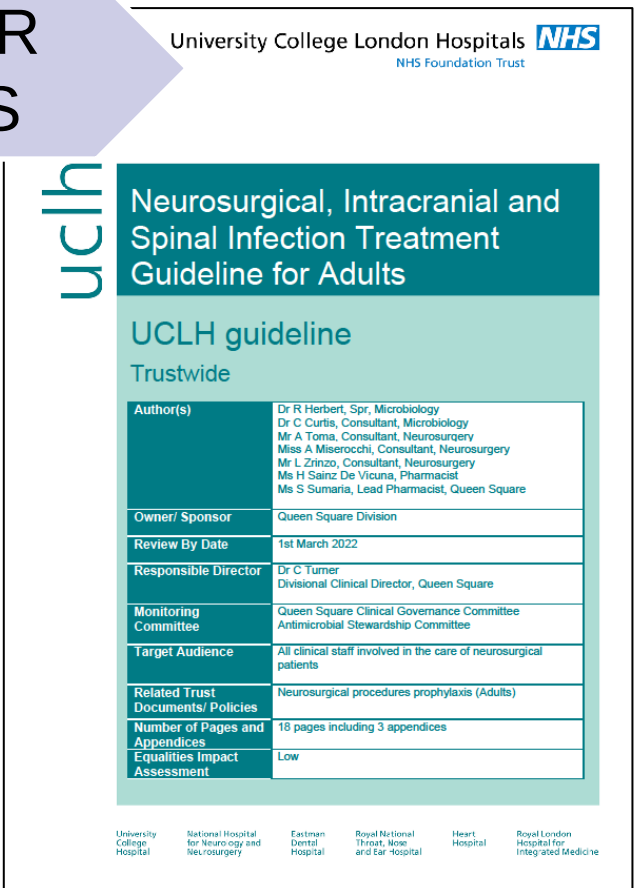
- If already on **PROPHYLACTIC LMWH**, LP should not be performed until 12h after dose
- Prophylactic LMWH should be delayed until 4h after LP performed
- Patients on **THERAPEUTIC LMWH** should not have LP until 24h after dose
- Patients on warfarin – LP should not be performed until INR is <1.4
- Patients on aspirin and other NSAIDs do not need LP delayed
- Patients on clopidogrel and DOACs (e.g. rivaroxaban, dabigatran) - LP delayed for 7 days - discuss with haematology
- Patients with known thrombocytopenia - LP should not be performed at platelet counts of <40 x 10⁹/L or with a rapidly falling platelet count
- LP should not be delayed for results of blood tests unless high clinical suspicion of bleeding diathesis

START ANTIBIOTIC + ANTIVIRAL TREATMENT AS SOON AS POSSIBLE

1st line: Ceftriaxone 2g IV BD + Aciclovir 10mg/kg (IBW if obese) TDS

If **BACTERIAL** cause suspected: ADD Dexamethasone 10mg IV qds - 1st dose given just before, with, or up to 12h after 1st dose of antibiotics

SEE SPECIFIC GUIDELINES FOR NEUROSURGICAL INFECTIONS



MENINGITIS – Regional infection		
Infection affects meninges, usually arachnoid, pia, and intervening space		
MENINGOCOCCAL	PNEUMOCOCCAL	TB
Ceftriaxone or Cefotaxime 2g IV BD for at least 5 days STERIODS If not pneumococcal meningitis, stop dexamethasone	Ceftriaxone or Cefotaxime 2g IV BD for 10-14 days STERIODS If pneumococcal meningitis confirmed, or thought probable (based on clinical, epidemiological and CSF parameters), continue dexamethasone for 4 days Consider areas with known risk of resistance (USA, South Africa, Spain) - may need vancomycin	Anti-TB drugs isoniazid / rifampin / pyrazinamide / + either streptomycin or ethambutol Steroids may be indicated Anti TB drugs only given when definitive diagnosis of TB made or after excluding other causes in comatose patients since once started these drugs must be given for a long time
Close contacts in 2 days prior to admission should be screened and given rifampicin for 2 days Possibly also staff involved in intubating patient	Screen for primary source of infection	Screen for primary focus of TB
Infection precautions Droplet propagated Normal precautions after 1 st dose of antibiotics given Use visor and closed suction		Infection precautions Infected patients require isolation precautions for 2 weeks from start of treatment

ENCEPHALITIS – Diffuse infection		
brain tissue +/- inflammation of meninges		
VIRAL ENTEROVIRUS & HSV-2	HSV-1 / VZV / ENTEROVIRUS	BACTERIAL
Symptomatic relief only and supportive therapy	Anti-viral therapy ACICLOVIR 10mg/kg (IBW if obese) for 21 days	Appropriate antibiotic Most common cause <i>Listeria</i> - Treat with amoxicillin

BRAIN ABSCESS – Focal infection

Intracerebral or subdural empyema

Surgery for lesions >2.5cm

Antibiotics
MEROPENEM 2g IV tds (review within 72h with culture results)
Abx treatment usually for 6-8 weeks

Steroids